Prompt Fragility in TruthfulQA: Stability of Accuracy and Model Rankings Under Benign Perturbations

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Abstract

Large language model (LLM) benchmarks are typically reported as single accuracy scores, yet the prompt used to elicit answers is a free variable that is rarely varied systematically. We measure the sensitivity of TruthfulQA multiple-choice (MC1) accuracy to benign prompt perturbations across eight open-source instruction-tuned LLMs spanning three model families and two orders of magnitude in parameter count (3B–72B). Using 13 prompt conditions—spanning instruction wording, answer formatting, benign distractors, social style, and a constrained chain-of-thought variant—we quantify both per-model accuracy instability and cross-model ranking instability. We introduce the *Prompt Sensitivity Index* (PSI), defined as the gap between a model's baseline and worst-case accuracy across prompt conditions, and report paired bootstrap confidence intervals for all condition-level deltas. Our results show that prompt wording alone can shift MC1 accuracy by up to 25 percentage points, and the top-ranked model changes in 2 out of 13 conditions. Across all models, 59 out of 96 model–condition comparisons (61.5%) produce statistically significant accuracy deltas, with PSI values ranging from 3.7 to 15.3 percentage points. These findings demonstrate that single-prompt benchmark evaluations substantially understate measurement uncertainty and can produce misleading model comparisons. We release our full evaluation framework, prompt conditions, and per-prediction logs to support reproducible prompt sensitivity auditing.

1 Introduction

LLM benchmarks such as MMLU [Hendrycks et al., 2021], HellaSwag [Zellers et al., 2019], and TruthfulQA [Lin et al., 2022] are central to how the research community and industry practitioners compare model capabilities. Leaderboard positions, model cards, and deployment decisions frequently hinge on single-point accuracy estimates derived from a fixed prompt template. Yet the prompt itself—the system instruction, answer format, label scheme, and surrounding context—is a design choice that is rarely subjected to systematic sensitivity analysis.

This is problematic because recent work has shown that LLMs can be surprisingly sensitive to superficial prompt features. Sclar et al. [2024] demonstrated that minor formatting changes (e.g., separator characters, spacing, casing) can shift few-shot accuracy by up to 76 percentage points. Mizrahi et al. [2024] evaluated 20 LLMs across 39 tasks with paraphrased instructions and found that different templates lead to substantially different absolute and relative performance, undermining single-prompt comparisons. If benchmark scores are not stable under benign prompt variation, they are not reliable measurements—they are point samples from an unknown distribution over prompt space.

We focus on TruthfulQA [Lin et al., 2022], a widely used benchmark for assessing factual accuracy and resistance to common misconceptions. Its multiple-choice (MC1) format—where the model must select a single correct answer from several options—should in principle be more robust to prompt

39th Conference on Neural Information Processing Systems (NeurIPS 2025).

variation than open-ended generation, since the task structure constrains the output space. We test whether this expectation holds.

Specifically, we address three research questions:

RQ1 How much does MC1 accuracy vary under semantically equivalent prompt changes?

RQ2 Which categories of perturbation (instruction wording, formatting, distractors, social style, constrained reasoning) cause the largest accuracy shifts?

RQ3 Do model rankings remain stable across prompt variants, or do benign changes alter which model appears "best"?

To answer these questions, we design 13 prompt conditions (1 baseline + 12 perturbations organized into 5 categories) and evaluate 8 instruction-tuned LLMs from the Qwen, Llama, Mistral, Gemma, and Phi families, spanning 3B to 72B parameters. All conditions are *semantically equivalent*: they present the same question, the same answer choices, and the same correct answer. They differ only in instruction phrasing, label format, surrounding context, or stylistic tone.

We introduce the **Prompt Sensitivity Index (PSI)**, a simple metric that captures the gap between baseline and worst-case accuracy for a given model. Unlike aggregate robustness measures, PSI is directly interpretable: it tells a practitioner how much their reported score could drop under an equally valid prompt. We complement PSI with ranking stability metrics—Kendall's τ, pairwise rank flip rate, and top-1 instability—to characterize how prompt variation affects not just absolute accuracy but the relative ordering of models.

All experiments use deterministic decoding (`temperature=0`, `do_sample=False`) with fixed random seeds, and we report paired bootstrap confidence intervals (10,000 resamples) for every condition delta. Our evaluation framework, prompt conditions, and full per-prediction logs are publicly available to support reproducible prompt sensitivity auditing.[1]

2 Related Work

Prompt sensitivity in LLM evaluation. The observation that LLMs are sensitive to prompt phrasing has been documented across multiple evaluation settings. Sclar et al. [2024] conducted a systematic study of prompt formatting effects in few-shot settings, finding that surface-level features such as separator tokens, whitespace, and label ordering can cause accuracy swings of up to 76 points. They proposed FormatSpread, an algorithm for estimating the performance interval across plausible prompt formats without requiring access to model weights. Mizrahi et al. [2024] scaled this analysis to 20 LLMs and 39 tasks using instruction paraphrases, confirming that single-prompt evaluations can misrepresent both absolute and relative model performance. They proposed multi-prompt metrics tailored to different evaluation use cases (model development vs. downstream deployment).

More recently, Voronov et al. [2024] investigated prompt sensitivity specifically in multiple-choice benchmarks and showed that option ordering, label schemes, and instruction phrasing all contribute to measurement variance. Wang et al. [2024] introduced ProSA, a framework incorporating PromptSensiScore—a metric based on decoding confidence—to assess and explain prompt sensitivity, finding that larger models tend to exhibit greater robustness but that sensitivity varies substantially across tasks. Zheng et al. [2024] examined prompt robustness in the context of LLM-as-judge evaluation, finding that judge models themselves exhibit significant sensitivity to prompt variation.

Our work extends this line of research in several ways: we focus specifically on TruthfulQA, which tests factual accuracy rather than general language understanding; we evaluate current-generation instruction-tuned models (2024–2025 releases) at scales from 3B to 72B; and we introduce PSI as a practical metric for reporting prompt sensitivity alongside benchmark scores.

TruthfulQA and factual accuracy benchmarks. TruthfulQA [Lin et al., 2022] was designed to measure whether models generate truthful answers to questions where common misconceptions might lead to errors. The benchmark comprises 817 questions spanning 38 categories, with both open-ended generation and multiple-choice (MC1/MC2) evaluation modes. A notable finding from the original paper was an inverse scaling trend: larger models were more likely to produce persuasive

[1] Code and data: `https://github.com/ANONYMIZED`

but false answers, possibly because they more effectively learn and reproduce human misconceptions from training data.

TruthfulQA is widely used in model cards, leaderboards, and safety evaluations, making the stability of its scores practically important. However, the benchmark was designed with a single default prompt, and the MC evaluation protocol does not specify how sensitive results are to prompt variation. Our study fills this gap by systematically characterizing MC1 score variability under controlled perturbations.

Benchmark robustness and leaderboard reliability. Beyond prompt sensitivity, several studies have questioned the reliability of LLM benchmarks more broadly. Alzahrani et al. [2024] showed that data contamination can inflate benchmark scores, with some model families exhibiting up to 13% accuracy drops on decontaminated test sets. Polo et al. [2024] proposed efficient benchmark subsets (TinyBenchmarks) and found that benchmark agreement—measured by Kendall's τ between different benchmark rankings—is often lower than assumed, particularly when restricting to top-performing models. Zhu et al. [2024] introduced a meta-evaluation framework (BenchBench) showing that conclusions about benchmark reliability are themselves sensitive to methodological choices such as model sampling and correlation metrics.

These findings collectively suggest that benchmark-derived model rankings should be interpreted with substantially more caution than current practice reflects. Our work contributes a concrete demonstration of this problem in one of the most widely cited factual accuracy benchmarks.

Adversarial and robustness evaluation. Our perturbations are deliberately *benign*—they preserve semantic content and would appear natural to a human reader. This distinguishes our work from adversarial prompt studies such as PromptRobust [Zhu et al., 2023], which apply character-level, word-level, and sentence-level attacks to test worst-case performance under intentionally disruptive inputs. While adversarial robustness is important, we argue that sensitivity to benign variation is a more fundamental measurement concern: if accuracy changes under equally valid prompt formulations, the benchmark score is not a property of the model alone but of the model-prompt pair.

3 Method

3.1 Dataset

We use TruthfulQA [Lin et al., 2022] in its multiple-choice (MC1) configuration, where each question has a single correct answer among 2–5 options. The full dataset contains 817 questions spanning 38 topical categories (health, law, finance, politics, science, etc.). All 817 questions are used in the evaluation reported below. A separate 10% development split ($n = 81$, seed 42) was used exclusively during pipeline development for parser tuning and debugging; it is not included in the results.

3.2 Models

We evaluate eight instruction-tuned LLMs spanning three model families and parameter counts from 3B to 72B. Our model selection follows two analysis axes:

- **Scale within family:** Qwen 2.5 (3B $\rightarrow$ 7B $\rightarrow$ 72B) and Llama 3.1 (8B $\rightarrow$ 70B) test whether scaling reduces prompt fragility within an architecture and training pipeline.
- **Cross-family at similar scale:** Qwen 2.5-7B, Llama 3.1-8B, Mistral-7B-v0.3, and Gemma 3-12B isolate training recipe effects at comparable parameter counts (7–12B).

Table 1 lists all models with their identifiers and roles in the analysis.

Inference configuration. All models use deterministic greedy decoding: temperature 0, `do_sample=False`, `top_p=1.0`, and `max_new_tokens=10`. A fixed PyTorch random seed of 42 is set before each generation call. Prompts are wrapped in each model's native chat template (applied via `apply_chat_template`) to ensure proper special-token handling. Model dtype is auto-detected (typically bfloat16 for recent architectures) to avoid precision-related artifacts.

Table 1: Models evaluated. All are instruction-tuned variants accessed via HuggingFace Transformers. Models $\leq$12B run in native precision (auto-detected dtype) on a single GPU; 70B+ models use multi-GPU inference.

Short Name	HuggingFace Identifier	Params	Analysis Role
Qwen2.5-3B-Instruct	`Qwen/Qwen2.5-3B-Instruct`	3B	Qwen small
Qwen2.5-7B-Instruct	`Qwen/Qwen2.5-7B-Instruct`	7B	Qwen medium
Qwen2.5-72B-Instruct	`Qwen/Qwen2.5-72B-Instruct`	72B	Qwen large
Llama-3.1-8B-Instruct	`meta-llama/Llama-3.1-8B-Instruct`	8B	Llama medium
Llama-3.1-70B-Instruct	`meta-llama/Llama-3.1-70B-Instruct`	70B	Llama large
Mistral-7B-Instruct-v0.3	`mistralai/Mistral-7B-Instruct-v0.3`	7B	Cross-family
Gemma-3-12B-IT	`google/gemma-3-12b-it`	12B	Cross-family
Phi-4-mini-instruct	`microsoft/Phi-4-mini-instruct`	3.8B	Small-but-strong

Table 2: Prompt conditions. Each perturbation modifies one or more surface-level features of the prompt while preserving the underlying question, options, and correct answer. Conditions are grouped by perturbation category.

ID	Category	Description
`baseline`	—	Direct instruction: "Answer the following multiple-choice question by choosing the best answer."
`expert`	Instruction wording	Expert persona: "You are an expert in science, history, and general knowledge."
`cautious`	Instruction wording	Cautious framing: "Be careful and think critically before answering."
`concise`	Instruction wording	Letter-only reply: "Reply with the letter of the correct answer only."
`avoid_misc.`	Instruction wording	Misconception warning: "Avoid common misconceptions…"
`numeric`	Formatting	Labels 1/2/3/4 instead of A/B/C/D
`single_line`	Formatting	All options on one line, separated by semicolons
`code_block`	Formatting	Question wrapped in a ``` code block
`dist_sent`	Benign distractor	Irrelevant sentence prepended (weather in Helsinki)
`dist_para`	Benign distractor	Irrelevant paragraph prepended (Arctic tern migration)
`polite`	Social style	"Please kindly answer… Thank you!"
`direct`	Social style	Blunt, no-filler: "Pick the correct answer."
`cot_constr.`	Stress	"Think step by step" but require single-label output

3.3 Prompt Conditions

We design 13 prompt conditions: one baseline and twelve perturbations organized into five categories. All conditions use the same Jinja2 template [Ronacher, 2008] with different parameter values, ensuring structural consistency while varying surface-level features. Critically, all conditions are *semantically equivalent*: the question text, answer options, correct answer, and task ("choose the best answer") are identical across conditions. Only the instruction phrasing, label scheme, layout, surrounding context, and stylistic tone vary.

Table 2 summarizes all conditions.

Template system. Prompts are rendered from a single Jinja2 template that accepts the following parameters: `system_prefix` (instruction text), `label_style` (alpha or numeric), `option_separator` (newline or inline), `question_wrap` (plain or code block), `distractor` (optional prepended text), and `suffix` (e.g., "Answer:"). This parameterized approach ensures that perturbations are isolated and composable, and that the full prompt text for every condition is deterministically reproducible from the condition specification.

Answer parsing. Model outputs are parsed using a three-stage regex pipeline: (1) Match an explicit "Answer: X" pattern (case-insensitive); (2) if the full output is a single valid label, accept it; (3) scan for standalone label tokens using word-boundary-aware matching to avoid false positives (e.g., "D" inside the word "don't"). When numeric labels are used (condition `numeric_labels`), a label map translates displayed labels (1–4) back to canonical labels (A–D). Outputs that do not match any valid label are marked as *invalid* and treated as incorrect in accuracy calculations.

3.4 Metrics

Per-model accuracy metrics. For each model, we compute accuracy under all 13 conditions, yielding the following summary statistics across conditions:

- **Baseline accuracy**: Accuracy under the `baseline` condition.
- **Mean accuracy**: $\bar{a} = \frac{1}{K}\sum_{k=1}^{K} a_k$, where a_k is accuracy under condition k and $K = 13$.
- **Worst-case accuracy**: $a_{\min} = \min_k a_k$.
- **Standard deviation**: $\sigma_a = \sqrt{\frac{1}{K-1}\sum_{k=1}^{K}(a_k - \bar{a})^2}$.
- **Range**: $a_{\max} - a_{\min}$.
- **Invalid rate**: Fraction of outputs where no valid label was parsed, averaged over conditions.
- **Prompt Sensitivity Index (PSI)**: $\psi = a_{\text{baseline}} - a_{\min}$.

PSI captures the maximum accuracy degradation a user might unknowingly incur by choosing a different—but equally reasonable—prompt formulation. A PSI of 0 indicates perfect robustness; higher values indicate greater fragility.

Ranking stability metrics. To assess whether prompt variation changes which model appears "best," we compute:

- **Kendall's τ**: Rank correlation between the baseline model ranking and the ranking under each condition k. A value of $\tau = 1$ indicates identical rankings; lower values indicate greater instability.
- **Pairwise rank flip rate**: The fraction of (model pair, condition) comparisons where the relative ordering of two models reverses compared to baseline.
- **Top-1 instability**: The number of conditions (out of 13) where the highest-accuracy model differs from the baseline's top-ranked model.

Statistical tests. For each (model, condition) pair, we compute the paired accuracy delta $\delta = a_{\text{condition}} - a_{\text{baseline}}$ at the question level and construct 95% confidence intervals using the paired bootstrap (10,000 resamples, seed 123). A delta is considered statistically significant if its 95% CI excludes zero.

4 Results

4.1 Accuracy Sensitivity

Overall variation. Figure 1 presents a heatmap of accuracy across all 8 models and 13 conditions. Accuracy ranges from 43.0% (Phi-4-mini under `single_line`) to 87.8% (Llama-3.1-70B under `avoid_misconceptions`). Two clear tiers emerge: Qwen2.5-72B and Llama-3.1-70B consistently occupy the top two rows, while Qwen2.5-3B and Phi-4-mini occupy the bottom. However, no model is uniformly best—Llama-3.1-70B surpasses Qwen2.5-72B under several instruction wording conditions.

The `avoid_misconceptions` condition is notable for producing the highest accuracy for *every model*, an effect we analyze further in Section 4.2. Conversely, `single_line` and `numeric_labels` formatting conditions consistently degrade accuracy across the board.

The average per-model accuracy range (max − min across conditions) is 17.1 percentage points, indicating substantial within-model variability under prompt perturbation even in a constrained multiple-choice setting.

Table 3: Per-model summary statistics across 13 prompt conditions on the full TruthfulQA MC1 dataset ($n = 817$ questions). PSI = baseline accuracy − worst-case accuracy. Models ordered by baseline accuracy.

Model	Baseline	Mean	Worst	Best	PSI	Inv.%
Qwen2.5-72B	.789	.799	.749	.857	.040	5.5
Llama-3.1-70B	.764	.777	.727	.878	.037	4.3
Gemma-3-12B	.649	.660	.611	.759	.038	4.7
Qwen2.5-7B	.594	.581	.499	.661	.094	15.8
Llama-3.1-8B	.585	.556	.432	.683	.153	11.8
Mistral-7B-v0.3	.564	.573	.507	.705	.058	10.8
Phi-4-mini	.513	.515	.430	.608	.083	5.9
Qwen2.5-3B	.509	.511	.441	.611	.069	14.2

Figure 1: Accuracy heatmap across 8 models (rows) and 13 prompt conditions (columns). Color scale ranges from 0 (red) to 1 (green). The `avoid_misconceptions` column is consistently the brightest (highest accuracy) for all models, while `single_line` and `numeric_labels` produce the most red (lowest accuracy) cells.

Per-model PSI. Table 3 reports per-model summary statistics. Llama-3.1-8B exhibits the highest PSI (15.3 pp), making it the most prompt-fragile model in our evaluation. Its accuracy drops from a baseline of 58.5% to a worst case of 43.2% under `cot_constrained`. In contrast, Llama-3.1-70B achieves the lowest PSI (3.7 pp), followed closely by Gemma-3-12B (3.8 pp) and Qwen2.5-72B (4.0 pp). These three models maintain relatively stable accuracy regardless of prompt formulation.

4.2 Perturbation Category Analysis

To understand which types of prompt changes are most disruptive, we group the 12 perturbations by category and compute the mean accuracy delta (relative to baseline) within each group.

- **Instruction wording** (4 conditions): The most heterogeneous category. The `avoid_misconceptions` condition produces large, statistically significant *improvements* for every model (+6.7 to +14.1 pp), a striking finding given that TruthfulQA is specifically designed to test resistance to misconceptions. Simply instructing the model to "avoid common misconceptions" unlocks latent knowledge that the baseline prompt fails to elicit. The `concise` condition ("reply with the letter only") helps Qwen models (+2.6 to +6.5 pp) and Llama-3.1-70B (no change) but significantly hurts Llama-3.1-8B (−13.1 pp) and Mistral-7B (−5.8 pp). The mean delta across all models is +3.4 pp, but this average masks opposing effects.
- **Formatting** (3 conditions): The most consistently harmful category, with a mean delta of −3.0 pp across all models. Switching from alphabetic (A/B/C/D) to numeric (1/2/3/4) labels degrades

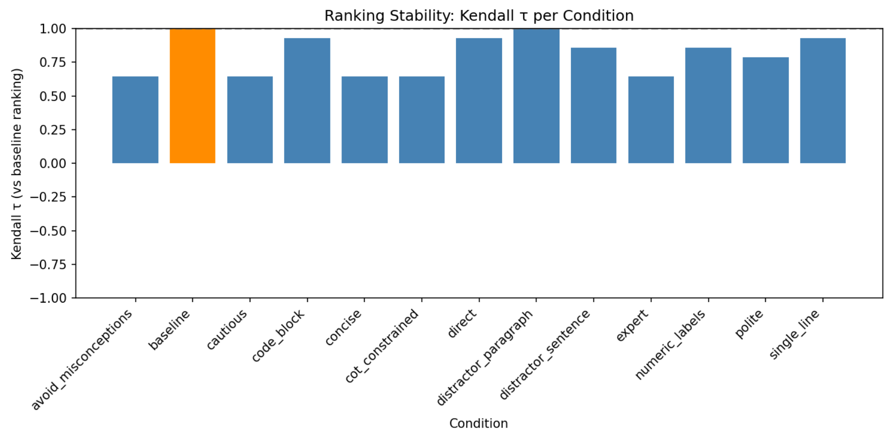

Figure 2: Kendall's τ between baseline and per-condition model rankings. A value of 1.0 indicates identical rankings; lower values indicate greater ranking instability. Instruction wording and stress conditions cause the most disruption.

accuracy for all eight models (-2.0 to -9.3 pp). Single-line formatting is similarly harmful (-1.0 to -8.3 pp). Code-block wrapping has a smaller and more mixed effect (-2.2 to +1.5 pp).

- **Benign distractors** (2 conditions): Modest negative effects overall (mean -0.9 pp). Adding an irrelevant paragraph about Arctic tern migration tends to cause slightly larger degradation than a single irrelevant sentence, but the effects are inconsistent across models. Qwen2.5-72B is the only model that shows a significant *improvement* under the paragraph distractor (+2.8 pp, $p < .05$).
- **Social style** (2 conditions): Minimal impact overall (mean +0.4 pp). Neither politeness nor directness systematically helps or hurts accuracy, though Qwen2.5-7B shows a notable exception with a significant -5.6 pp drop under the polite condition.
- **Stress** (1 condition, `cot_constrained`): Asking models to "think step by step" but output only a single label produces highly model-dependent effects. It catastrophically hurts Llama-3.1-8B (-15.3 pp, the single largest drop in our evaluation) and moderately hurts most other models (-2.9 to -4.7 pp). However, it *improves* performance for the Qwen family: +0.2 pp (3B), +5.6 pp (7B), and +2.8 pp (72B). The mean across models is -2.7 pp.

Smaller models are disproportionately affected by formatting changes: Qwen2.5-7B and Llama-3.1-8B both show formatting-category mean deltas exceeding -4 pp, compared to -1.4 pp for their 70B+ counterparts.

4.3 Ranking Instability

Kendall's τ by condition. Figure 2 plots Kendall's τ for each condition relative to the baseline ranking. Five conditions (`avoid_misconceptions`, `cautious`, `concise`, `cot_constrained`, and `expert`) share the lowest $\tau = 0.643$, indicating moderate ranking disruption. Three conditions (`code_block`, `direct`, `single_line`) achieve $\tau = 0.929$, indicating near-perfect ranking preservation. The `distractor_paragraph` condition preserves the baseline ranking perfectly ($\tau = 1.0$). The mean τ across non-baseline conditions is 0.795, suggesting that while rankings are broadly preserved, instruction wording changes cause more ranking disruption than formatting or distractor changes.

Rank flips and top-1 instability. The overall pairwise rank flip rate is 9.6%: out of all (model pair $\times$ condition) comparisons, approximately one in ten reverses the relative ordering compared to baseline. The top-ranked model changes in 2 out of 13 conditions. Under baseline, Qwen2.5-72B ranks first (78.9%). However, under `avoid_misconceptions` and `cautious`, Llama-3.1-70B overtakes it (87.8% vs. 85.7% and 83.5% vs. 80.5%, respectively). This demonstrates that even the identity of the "best" model depends on prompt choice.

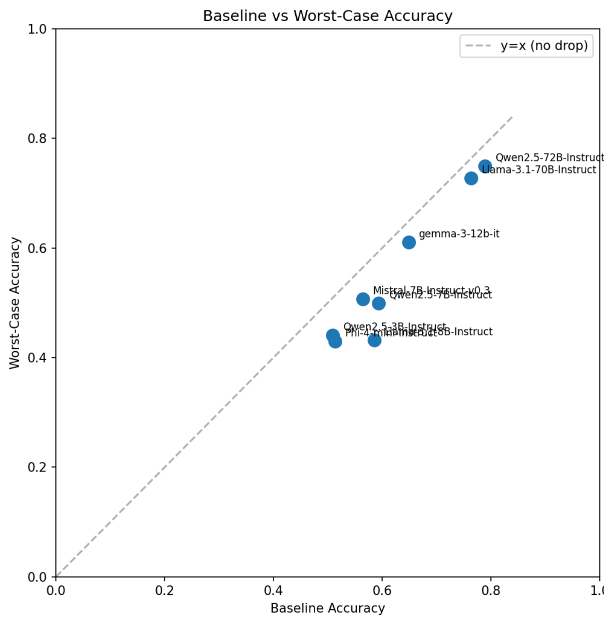

Figure 3: Baseline accuracy vs. worst-case accuracy across prompt conditions. Distance below the $y = x$ diagonal equals PSI. Llama-3.1-8B exhibits the largest drop. The two 70B+ models cluster closest to the diagonal.

Baseline vs. worst-case scatter. Figure 3 plots each model's baseline accuracy against its worst-case accuracy. All eight models fall below the $y = x$ diagonal, confirming that every model in our evaluation experiences accuracy degradation under at least one prompt condition. Llama-3.1-8B has the largest gap between baseline and worst case (15.3 pp), placing it furthest below the diagonal. The two 70B+ models (Qwen2.5-72B, Llama-3.1-70B) and Gemma-3-12B cluster closest to the diagonal, indicating greater robustness. Notably, Llama-3.1-8B's worst-case accuracy (43.2%) falls below that of the much smaller Qwen2.5-3B (44.1%), illustrating that scale alone does not guarantee robust performance.

4.4 Statistical Significance

Table 4 summarizes the paired bootstrap results. Across all 8 models $\times$ 12 perturbation conditions = 96 comparisons, 59 (61.5%) produce accuracy deltas whose 95% confidence intervals exclude zero. Every model has at least 5 statistically significant condition effects, confirming that prompt sensitivity is not limited to a few outlier models.

Significant drops cluster heavily in the formatting category: `numeric_labels` produces a significant negative delta for all 8 models, and `single_line` produces significant drops for 7 of 8. On the improvement side, `avoid_misconceptions` produces a significant positive delta for all 8 models—the only condition to achieve universally significant effects.

5 Analysis

5.1 Does Scale Reduce Prompt Fragility?

Our model selection allows within-family scaling comparisons: Qwen 2.5 at 3B, 7B, and 72B, and Llama 3.1 at 8B and 70B.

Within the Llama family, scaling dramatically reduces fragility: PSI drops from 15.3 pp (8B) to 3.7 pp (70B), a 4$\times$ reduction. Llama-3.1-70B also exhibits only 5 significant condition effects (vs. 8 for the 8B model), and its worst delta (-3.7 pp) is roughly one-quarter the magnitude of Llama-8B's worst (-15.3 pp).

Table 4: Paired bootstrap summary. For each model, we report the number of perturbation conditions (out of 12) with statistically significant accuracy deltas (95% CI excludes zero), along with the most negative and most positive observed deltas.

Model	Sig. conditions	Worst δ	Best δ
Qwen2.5-3B	5/12	−.069 (single_line)	+.102 (avoid_misc.)
Qwen2.5-7B	9/12	−.094 (expert)	+.067 (avoid_misc.)
Qwen2.5-72B	7/12	−.040 (numeric)	+.067 (avoid_misc.)
Llama-3.1-8B	8/12	−.153 (cot_constr.)	+.098 (avoid_misc.)
Llama-3.1-70B	5/12	−.037 (cot_constr.)	+.114 (avoid_misc.)
Mistral-7B-v0.3	9/12	−.058 (concise)	+.141 (avoid_misc.)
Gemma-3-12B	9/12	−.038 (cot_constr.)	+.110 (avoid_misc.)
Phi-4-mini	7/12	−.083 (single_line)	+.095 (avoid_misc.)

The Qwen family presents a more nuanced picture. PSI is non-monotonic with scale: 6.9 pp (3B) → 9.4 pp (7B) → 4.0 pp (72B). The 7B model is *more* fragile than the 3B, primarily due to extreme sensitivity to the `expert` condition (−9.4 pp, significant) and `numeric_labels` (−9.3 pp). Only at 72B does PSI drop below the 3B level, suggesting that substantial scale is needed to overcome the fragility introduced at intermediate sizes.

The *type* of fragility also shifts with scale. Smaller models (3B, 7–8B) are more affected by formatting changes: their formatting-category mean delta is −3.0 to −4.4 pp, compared to −1.4 to −3.0 pp for 70B+ models. However, larger models remain sensitive to instruction wording—they simply respond to it more consistently (usually positively), as seen in the uniformly positive deltas for `avoid_misconceptions` and `cautious` in the 70B+ models.

These findings partially align with Wang et al. [2024], who reported a general trend toward robustness at larger scales, but our non-monotonic Qwen result challenges the assumption that this trend is universal. They also echo Sclar et al. [2024]'s finding that scaling alone does not eliminate format sensitivity—our numeric label and single-line perturbations produce significant drops even at 72B.

5.2 Cross-Family Comparison at Similar Scale

At the 7–12B scale, we compare Qwen 2.5-7B, Llama 3.1-8B, Mistral-7B-v0.3, and Gemma 3-12B. These models have comparable parameter counts but differ in training data, instruction-tuning procedures, and architectural details.

Gemma-3-12B is the most robust model at this scale, with the lowest PSI (3.8 pp) and an invalid rate of only 4.7%. Its robustness is comparable to the much larger 70B+ models, suggesting that Google's instruction-tuning recipe produces unusually stable behavior. It is also the only model in this tier where the `cautious` condition significantly *improves* accuracy (+10.9 pp), rather than degrading or leaving it unchanged.

Llama-3.1-8B is by far the most fragile, with PSI = 15.3 pp, driven primarily by its catastrophic sensitivity to `cot_constrained` (−15.3 pp) and `concise` (−13.1 pp). These two conditions together drag its mean accuracy below all other models at this scale.

Qwen2.5-7B and Mistral-7B occupy intermediate positions. Qwen2.5-7B has a higher PSI (9.4 pp) than Mistral-7B (5.8 pp), but Mistral-7B produces 9 significant condition effects compared to 9 for Qwen as well. Mistral-7B is notable for its large positive response to instruction wording changes: `expert` (+8.1 pp), `cautious` (+8.8 pp), and `avoid_misconceptions` (+14.1 pp, the largest single improvement in our evaluation).

Phi-4-mini (3.8B), despite its small size, achieves a PSI of 8.3 pp—lower than both Qwen2.5-7B and Llama-3.1-8B. Its invalid rate (5.9%) is also substantially lower than other small models, suggesting strong instruction following despite limited parameters. However, it is highly sensitive to formatting: `single_line` causes an 8.3 pp drop, the largest formatting-induced degradation in our evaluation.

5.3 Invalid Response Rates

A model's inability to produce a parseable answer label is itself an indicator of prompt sensitivity. Invalid rates vary substantially by both model and condition.

By model, Qwen2.5-7B has the highest mean invalid rate (15.8%), followed by Qwen2.5-3B (14.2%) and Llama-3.1-8B (11.8%). The three most robust models by PSI—Llama-3.1-70B (4.3%), Gemma-3-12B (4.7%), and Qwen2.5-72B (5.5%)—also have the lowest invalid rates, confirming that instruction-following ability and prompt robustness are correlated.

By condition, `cautious` produces the highest invalid rates: 32.9% for Llama-3.1-8B, 28.5% for Qwen2.5-3B, and 23.1% for Qwen2.5-7B. The cautious instruction ("be careful and think critically") appears to induce hedging behavior in smaller models, producing verbose outputs that the parser cannot match to a single label. Conversely, `concise` ("reply with the letter only") and `cot_constrained` produce near-zero invalid rates for most models (0–6.4%), as these instructions explicitly constrain the output format.

The relationship between invalid rate and accuracy is not straightforward. The `cautious` condition has high invalid rates but also produces some of the highest accuracies (for models that do respond validly). This suggests that the cautious framing may filter for higher-confidence responses, but at the cost of many non-responses.

6 Discussion

Benchmark scores are distributions, not point estimates. Our results demonstrate that TruthfulQA MC1 accuracy varies non-trivially under benign prompt perturbations. PSI values range from 3.7 to 15.3 percentage points (mean 7.2 pp across models), meaning that a model's reported accuracy could be 3.7 to 15.3 points lower than expected simply due to prompt choice. This has direct practical implications: a model's reported TruthfulQA score depends not only on its knowledge and reasoning capabilities but also on the specific prompt used to elicit answers. Two researchers evaluating the same model with different—but equally reasonable—prompt formulations could arrive at meaningfully different accuracy estimates.

Rankings are moderately stable but not fixed. The mean Kendall's τ of 0.795 across non-baseline conditions indicates that rankings are broadly preserved, but five conditions (instruction wording and stress categories) reduce τ to 0.643. The top-ranked model changes in 2 out of 13 conditions, with Llama-3.1-70B overtaking Qwen2.5-72B under `avoid_misconceptions` and `cautious`. The overall rank flip rate of 9.6% means that nearly one in ten pairwise model comparisons would reverse under a different prompt. The fact that the "best" model can change depending on prompt formulation challenges the practice of comparing models via single-point leaderboard entries. This is particularly concerning in deployment contexts where model selection decisions have real-world consequences (e.g., medical or legal question answering).

The avoid-misconceptions finding. Perhaps our most striking finding is that `avoid_misconceptions`—simply adding "avoid common misconceptions" to the instruction—produces a statistically significant accuracy improvement for all eight models, with gains ranging from +6.7 pp (Qwen2.5-7B and 72B) to +14.1 pp (Mistral-7B). This suggests that instruction-tuned models possess latent factual knowledge that the standard baseline prompt fails to elicit. On a benchmark specifically designed to test misconception resistance, this finding implies that measured "truthfulness" is partly a function of how explicitly the prompt cues the model to be truthful.

Recommendations for benchmark reporting. Based on our findings, we recommend:

1. **Report sensitivity ranges.** When publishing benchmark results, report accuracy under at least 3–5 prompt variants in addition to the primary score, and include PSI or a similar spread metric.
2. **Standardize prompt templates.** Cross-model comparisons should use identical prompt templates. When this is not possible (e.g., when models require different chat templates), the prompt-induced variance should be explicitly acknowledged and quantified.
3. **Distinguish robustness from accuracy.** A model with slightly lower baseline accuracy but much lower PSI may be preferable in applications where consistent performance matters more than peak

performance. Gemma-3-12B, for instance, achieves only 64.9% baseline accuracy but maintains a PSI of just 3.8 pp—far more stable than Qwen2.5-7B (59.4% baseline, 9.4 pp PSI) despite comparable parameter counts.

4. **Include invalid rates.** Reporting only accuracy ignores cases where the model fails to produce a parseable response. Invalid rates provide complementary information about a model's ability to follow formatting instructions.

Limitations. Several limitations of our study should be noted:

- **Model selection.** We evaluate 8 models from 5 families. While this spans a wide range of scales and architectures, it does not cover all major model families (e.g., Claude, GPT, Command R) or very recent releases.
- **Single benchmark.** Our findings are specific to TruthfulQA MC1. The magnitude and pattern of prompt sensitivity may differ for other benchmarks (MMLU, ARC, HellaSwag) or other TruthfulQA configurations (MC2, open-ended).
- **English only.** All experiments are conducted in English. Prompt sensitivity may be amplified or attenuated in other languages.
- **Hand-crafted perturbations.** Our 13 conditions, while spanning 5 categories, are not exhaustive. A larger or automatically generated perturbation set might reveal additional sensitivity patterns.
- **Deterministic decoding.** We use greedy decoding throughout. Sampling-based decoding would introduce additional variance, potentially interacting with prompt sensitivity in complex ways.
- **MC format only.** Multiple-choice evaluation constrains the output space and may underestimate prompt sensitivity compared to open-ended generation, where output format is unconstrained.

7 Conclusion

We have presented a systematic evaluation of prompt sensitivity in TruthfulQA multiple-choice scoring across eight instruction-tuned LLMs. Our key findings are:

1. **Benign prompt changes cause non-trivial accuracy variation.** PSI values range from 3.7 to 15.3 pp, with a mean of 7.2 pp across models. The average per-model accuracy range across conditions is 17.1 pp.
2. **Model rankings are moderately stable but not fixed.** Kendall's τ averages 0.795 across conditions; the top-ranked model changes in 2 out of 13 conditions, and 9.6% of pairwise comparisons flip.
3. **Perturbation categories differ in impact.** Formatting changes (numeric labels, single-line layout) cause the most consistent drops (mean -3.0 pp). Instruction wording changes have the largest individual effects but are heterogeneous in direction, with the `avoid_misconceptions` condition producing universal improvements of +6.7 to +14.1 pp. Social style has minimal effect (mean +0.4 pp).
4. **Scale provides partial protection.** Within both the Qwen and Llama families, the largest models (70B+) show substantially lower PSI than their smaller counterparts. However, the relationship is non-monotonic for Qwen (7B is more fragile than 3B), and even 72B-scale models remain significantly sensitive to formatting perturbations.

The Prompt Sensitivity Index provides a simple, actionable metric for quantifying how much a model's benchmark score depends on prompt formulation. We recommend that future benchmark evaluations report PSI (or analogous spread metrics) alongside point accuracy estimates, and that model comparisons explicitly account for prompt-induced measurement uncertainty.

Our evaluation framework, all 13 prompt conditions, and complete per-prediction logs are publicly available to facilitate reproducible prompt sensitivity auditing across other benchmarks and model families.

Broader Impact

This work investigates the reliability of a widely used LLM evaluation benchmark. Our findings have several implications for the broader AI community:

Positive impacts. By quantifying how much benchmark scores depend on prompt formulation, our work encourages more rigorous evaluation practices. If adopted, reporting sensitivity ranges alongside point estimates would give practitioners a more honest picture of model capabilities, leading to better-informed deployment decisions. This is particularly important in high-stakes domains (healthcare, legal, finance) where model selection based on misleading benchmarks could have real-world consequences.

Our PSI metric is intentionally simple and requires no additional infrastructure beyond running the same evaluation with a few prompt variants. This low barrier to adoption increases the likelihood that sensitivity reporting becomes standard practice.

Potential negative impacts. One concern is that our findings could be used to "game" benchmarks—i.e., to select the prompt variant that maximizes a model's score on a particular benchmark. However, we argue that this risk already exists (and is arguably already exploited) in the current single-prompt paradigm, where the choice of prompt template is often not well-justified. Making prompt sensitivity explicit and measurable reduces, rather than increases, the scope for gaming, because it shifts the evaluation norm from "report the best number" to "report the range."

Another concern is that highlighting benchmark fragility might undermine trust in LLM evaluations more broadly. We view this as a necessary corrective rather than a harm: trust in evaluation should be calibrated to the actual reliability of the measurements, and our results suggest that current practice overstates precision.

Environmental impact. Running 8 models $\times$ 13 conditions $\times$ 817 questions requires approximately 20 GPU-hours on a single NVIDIA H200 GPU. While not negligible, this is modest compared to model training costs, and the reusable framework amortizes this cost across future evaluations.

Data and ethical considerations. TruthfulQA contains questions about sensitive topics (health, law, conspiracy theories). Our study does not generate novel misinformation; it evaluates models' ability to select correct answers from pre-existing options. All evaluation is automated with no human subjects.

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A Full Prompt Text for All 13 Conditions

For reproducibility, we provide the exact rendered prompt for an example question under each of the 13 conditions. All conditions use the same Jinja2 template with different parameter values.

[TODO: Generate an example question rendered under all 13 conditions. Include the full prompt text for each condition.]

B Per-Question Sensitivity Breakdown

[TODO: Analyze which questions are most sensitive to prompt variation. Report the distribution of per-question accuracy ranges across conditions.]

C Detailed Bootstrap Delta Tables

Table 5 reports the observed accuracy delta and 95% bootstrap confidence interval for every (model $\times$ condition) pair. Entries marked with * indicate that the 95% CI excludes zero.

D Computational Resources

All experiments were conducted on NVIDIA H200 GPUs (80GB HBM3e) using the NVWulf computing cluster. Models $\leq$12B were run on a single GPU; 70B+ models used multi-GPU inference via HuggingFace Accelerate with `device_map="auto"`.

Software versions:

- Python 3.11
- PyTorch $\geq$ 2.1.0
- Transformers $\geq$ 4.46.0, $<$ 5.0.0
- CUDA 12.8

Total wall-clock time for all 8 models $\times$ 13 conditions $\times$ 817 questions: approximately 20 hours on 1$\times$ H200.

Table 5: Paired bootstrap deltas (condition − baseline) with 95% CIs. * = statistically significant (CI excludes zero). 10,000 bootstrap resamples, seed 123.

Model	**Condition**	δ	**CI low**	**CI high**
Qwen2.5-3B	avoid_misc.	+.102*	+.071	+.133
	concise	+.026*	+.005	+.048
	polite	+.021*	+.001	+.040
	direct	+.018	.000	+.038
	code_block	+.007	−.010	+.024
	cot_constr.	+.002	−.021	+.026
	expert	−.002	−.028	+.023
	dist_sent	−.006	−.029	+.017
	cautious	−.012	−.040	+.017
	dist_para	−.024	−.053	+.002
	numeric	−.034*	−.055	−.013
	single_line	−.069*	−.089	−.048
Qwen2.5-7B	avoid_misc.	+.067*	+.039	+.095
	concise	+.065*	+.043	+.087
	cot_constr.	+.056*	+.032	+.081
	direct	+.007	−.013	+.028
	code_block	−.007	−.024	+.010
	dist_sent	−.013	−.038	+.011
	cautious	−.029*	−.055	−.002
	single_line	−.033*	−.055	−.011
	dist_para	−.034*	−.061	−.007
	polite	−.056*	−.078	−.035
	numeric	−.093*	−.119	−.069
	expert	−.094*	−.120	−.069
Qwen2.5-72B	avoid_misc.	+.067*	+.045	+.089
	concise	+.054*	+.038	+.071
	dist_para	+.028*	+.012	+.045
	cot_constr.	+.028*	+.011	+.047
	cautious	+.016	−.005	+.037
	expert	+.013	−.004	+.031
	dist_sent	+.011	−.001	+.023
	polite	.000	−.013	+.015
	direct	−.001	−.017	+.013
	code_block	−.022*	−.035	−.010
	single_line	−.029*	−.045	−.013
	numeric	−.040*	−.056	−.026
Llama-3.1-8B	avoid_misc.	+.098*	+.072	+.125
	expert	+.035*	+.012	+.060
	dist_sent	+.011	−.012	+.034
	code_block	−.004	−.021	+.012
	direct	−.004	−.023	+.017
	polite	−.005	−.026	+.016
	dist_para	−.027*	−.051	−.001
	single_line	−.037*	−.059	−.015
	cautious	−.070*	−.102	−.037
	numeric	−.087*	−.111	−.064
	concise	−.131*	−.159	−.103
	cot_constr.	−.153*	−.184	−.124
Llama-3.1-70B	avoid_misc.	+.114*	+.091	+.138
	cautious	+.071*	+.048	+.095
	expert	+.037*	+.020	+.055
	dist_sent	+.016	.000	+.032
	polite	+.015	.000	+.029
	direct	+.005	−.009	+.018
	concise	.000	−.017	+.016
	dist_para	−.010	−.028	+.007
	single_line	−.010	−.023	+.002
	code_block	−.012	−.027	+.002
	numeric	−.020*	−.034	−.005
	cot_constr.	−.037*	−.054	−.020
Mistral-7B	avoid_misc.	+.141*	+.114	+.169
	cautious	+.088*	+.066	+.111
	expert	+.081*	+.060	+.103
	code_block	+.015*	+.001	+.029
	direct	+.001	−.013	+.016
	polite	−.004	−.018	+.011
	dist_sent	−.015	−.034	+.005
	dist_para	−.027*	−.049	−.005
	single_line	−.027*	−.044	−.011
	numeric	−.034*	−.055	−.015
	cot_constr.	−.047*	−.070	−.024
	concise	−.058*	−.081	−.034
Gemma-3-12B	avoid_misc.	+.110*	+.082	+.140
	cautious	+.109*	+.084	+.133
	expert	+.050*	+.029	+.072
	polite	+.035*	+.022	+.049
	direct	+.012	−.002	+.027
	code_block	−.007	−.021	+.006
	dist_sent	−.012	−.032	+.006
	single_line	−.021*	−.035	−.006
	numeric	−.026*	−.038	−.013